Propagation of Ornamental Plants
Vol. 00, № 0, 2000: 00-00

AXILLARY SHOOTS INDUCTION AND RHIZOGENESIS IN CURLY BIRCH UNDER A SHORTENED AND AN EXTENDED LIGHT-DARK CYCLES

Lidiya Vetchinnikova[1*], Tatjana Shibaeva[2], Kseniya Gudkova[1], and Alexander Titov[2]

[1]Forest Research Institute, Karelian Research Center, Russian Academy of Sciences,
11 Pushkinskaya str., 185910 Petrozavodsk, Russia, *E-mail: vetchin@krc.karelia.ru
[2]Institute of Biology, Karelian Research Center, Russian Academy of Sciences,
11 Pushkinskaya str., 185910 Petrozavodsk, Russia

Abstract

The article reports a study of the effects of a shortened (8/4 h, light/dark) and an extended (48/24 h) light-dark cycles (LDC) on growth of axillary shoots and rhizogenesis in curly birch *Betula pendula* Roth var. *carelica* (Mercklin) Hämet-Ahti *in vitro*, as well as their *ex vitro* aftereffects. Under the 8/4 h LDC shoot biomass was reduced (versus the 16/8 h control) as a result of growth inhibition of new axillary shoots and leaves. In the 48/24 h LDC, on the contrary, the length of the original shoots and the stem biomass increased (including for newly formed axillary shoots). The stimulating effect of the extended LDC was seen also during rhizogenesis, with manifestations being earlier root formation, higher number of roots and their larger size. The differences in growth parameters observed during axillary shoots formation and rhizogenesis *in vitro* under non-standard LDCs persisted in the regenerated plants for one month after their transfer to standard light-dark conditions *ex vitro*. Shortened and extended LDCs inhibited callus formation at the shoot base. The findings suggest that the extended 48/24 h LDC can be used to stimulate axillary shoot emergence and development, thereby enhancing clonal micropropagation performance. Conversely, the shortened 8/4 h LDC restrains shoot formation and thus reduces the frequency of subculturing (and the associated labor costs) during the off-season. Thus, abnormal LDCs differing from the standard 24-hour cycle can be used to handle specific tasks in cultivating the valuable curly birch *in vitro* more efficiently while enabling the application of energy-saving regimes to reduce the overall costs of the operations.

Key words: abnormal light-dark cycle, *Betula pendula*, growth, development, *in vitro* culture

PROOF

INTRODUCTION

Among all abiotic factors that follow certain patterns during the day and especially seasonally, the most critical for plant life are temperature and light (Maurya and Bhalerao 2017, Chu et al. 2024). In natural conditions, temperature and light signals primarily regulate the growth and development of plants, especially perennials. It is well-known that in deciduous woody species, both temperature and daylight duration significantly affect the seasonal patterns of growth and development over the annual cycle (Azeez and Sane 2015, Singh et al. 2021). In autumn, responding to shorter daylight and lower temperatures, woody plants cease to grow and fall into dormancy (Puhakainen et al. 2004, Junttila and Hänninen 2012), while in spring, as days grow longer and temperatures rise, growth and development are resumed (Singh et al. 2021). The light impact is particularly pronounced at mid- and high latitudes, where photoperiods range from 24/0 h ("polar day") to 0/24 h ("polar night"), and these changes are critically consequential to plant life, putting them in the focus of researchers' attention (Olsen et al. 2010, Skinner et al. 2018, Singh et al. 2021).

It is worth noting that in recent decades, the number of publications on the effect of light on the growth and development of cultivated plants (Dutta Gupta and Agarwal 2017, Janda et al. 2021) has increased significantly, largely due to the wide adoption of light-emitting diode (LED) technologies, which are increasingly used in greenhouse production of planting stock and in controlled-environment horticulture and floriculture. Essentially, LEDs have evolved into an alternative light source offering advantages over traditional gas-discharge or fluorescent lamps, such as adjustable levels of photosynthetically active radiation (PAR) and spectral composition for optimizing photosynthesis rate and plant growth, small size, longer service life, and

Received: June 15, 2025

Accepted: July 18, 2025

lower heat output (Dutta Gupta and Agarwal 2017).

Light acts as an important environmental factor during the cultivation of isolated tissues and organs. Although the effect of light on photosynthesis *in vitro* is less significant than in plants *in vivo* (Economou and Read 1987), its regulatory role is nevertheless very important under *in vitro* conditions. The nature and quantitative effect of light on plant growth and development depends on the spectral composition, intensity and duration of illumination. Studies researching light effects typically focus on its intensity and/or spectral composition (Pinker et al. 1989, Sæbø et al. 1995, Zhao et al. 2020, Kondratovičs and Zeps 2022), mostly under the same standard photoperiod (e.g., 16/8 h). However, the effect of the photoperiod per se, especially non-standard light-dark cycles (LDC) other than the 24-h cycle, is much less explored (Wynne and McDonald 2002, Hartmann et al. 2002, Vaillant et al. 2005, Mogilevskaya and Zholobova 2025). In particular, there are not many experimental studies on the effects of shortened and/or extended light-dark cycles on the growth and development of deciduous woody plants *in vitro* (Welander 1988, Morini et al. 1990, 1992, 1993, Zimmerman and Scorza 1994, Morini and Perrone 2006).

Besides the purely scientific interest it provokes, the research in this field could have also obvious practical implications. This could be particularly important considering the opportunities arising for mass-production of planting stock of economically valuable species. A notable example is the curly birch, *Betula pendula* Roth var. *carelica* (Mercklin) Hämet-Ahti, known for its highly ornamental and hard figured wood (Vetchinnikova and Titov 2021). Due to the diversity of growth forms of curly birch, it is used in urban plantings and landscape design.

In view of the above, our research aimed to study axillary shoots induction and rhizogenesis in curly birch (*Betula pendula* var. *carelica*) *in vitro* under a shortened and an extended light-dark cycles.

MATERIALS AND METHODS

The study object was a curly birch *Betula pendula* var. *carelica* clone from an *in vitro* collection (Collection 2025). *In vitro* originated axillary shoots with a length 0.5 cm and one or two ca. 1 × 2 mm leaf blades were used as initial explants. They were cultivated on Murashige and Skoog (1962) (MS) medium supplemented with 2.5% sucrose and 0.5 mg l^{-1} 6-benzylaminopurine (BAP) for axillary shoots induction. The number of newly formed axillary shoots, number of leaves, elongation and wet weight of shoots, and callus were measured after 4 weeks. For the induction of rhizogenesis, axillary shoots with length 2.0-2.5 cm and three to five leaf blades were used. They were cultivated on half-strength MS medium supplemented with 2.5% sucrose and 0.1 mg l^{-1} indole-3-butyric acid (IBA) (Wynne and McDonald 2002). The rooted shoots (%), the number, and length of the roots (mm) were determined after two weeks.

Shoots were cultured in test tubes (16 mm diameter) in Biobase BJPX-RG-P160c climate chambers (China) where the temperature (26 ± 1°C) and relative air humidity (30 ± 2%) during the experiments were constant. Induction of axillary shoots and rhizogenesis were initiated under the shortened (8/4 h day/night) and the extended (48/24 h) LDCs and photosynthetic photon flux density (PPFD) of 56 μmol m^{-2} s^{-1}. The control was the 16/8 h photoperiod, regarded as the most favorable for shoot cultivation and rhizogenesis from shoots *in vitro* (Vetchinnikova et al. 2017).

Five replications of each treatment were used and each contained 10-30 axillary shoots. The *in vitro* experiments were evaluated after 26 days for growth of axillary shoots and 14 days for rhizogenesis.

To reveal possible aftereffects of different LDCs (8/4, 16/8, and 48/24 h), the rooted plants (15 plants from each variant) were transferred from *in vitro* to *ex vitro* conditions. They were acclimatized for two weeks into standard mini (0.44 × 0.22 × 0.13 m) lid-covered greenhouses (18 cells in 3 trays), where humidity was maintained at 80-90% with regular ventilation (Fig. 1A). Then the plants were transferred for 6 weeks to cylindrical plastic containers (first of 250 ml volume, then 3 l) filled with the same substrate as the mini greenhouses (peat : vermiculite 3 : 1, v : v) and placed on shelves in a space with identical artificial PPDF (100 μmol m^{-2} s^{-1}), 16/8 h photoperiod, 23/21 ± 2°C, and with relative air humidity (60-70%) maintained by a BONECO 7135 ultrasonic humidifier (Switzerland). Three replications, each containing 15 plantlets were used for each treatment.

The aftereffects of shortened and extended LDCs on *ex vitro* growth of regenerated plants were evaluated regulary from the 15th to the 90th day, based on their length.

Data analyses was performed in Microsoft Excel Office 2108 (Build 14334.20136). For original shoot length, original shoot elongation, number of leaves, callus mass, stem mass, leaf mass, total biomass, and number of roots, one-way analysis of variance (ANOVA) was applied, followed by Tukey's HSD post-hoc test. For shoot mass, number of shoots and root length, the non-parametric Kruskal-Wallis test was used, with subsequent pairwise comparisons via the Mann-Whitney U test. For the aftereffect analysis, at each time point, one-way ANOVA was performed to compare the effects of different light-dark cycles, followed by Tukey's HSD post-hoc test for multiple comparisons. Statistical significance was determined at $p \leq 0.05$.

RESULTS

Growth of axillary shoots in curly birch under shortened and extended LDCs

Abnormal LDCs differing from the standard (16/8 h) photoperiod significantly affected axillary shoots formation cultivated *in vitro* (Fig. 1). The shoot mass in the shortened LDC (8/4 h) treatment was lower than in the control because of a reduction in the number of new axillary shoots and leaves (by a factor of 1.2 and 1.4 times, respectively). However, the elongation and total length of the initial shoot and the stem mass remained close to the control. Total biomass (shoots and callus) under the 8/4 h LDC was about 115.4 mg (ranging from 87 to 167 mg), which was nearly 30% lower than that of the control with 161.8 mg on average (Table 1).

In experiments with an extended LDC (48/24 h), on the contrary, an increase was observed both in the length of initial shoots (by 30%) and in stem mass, including newly formed axillary shoots (by 50%). The number of shoots and leaves in this variant was nearly the same as in the control, although total leaf mass was significantly lower (1.4 times) (Table 1).

The callus mass in the experimental LDC treatments was lower by over 25% than in the control (Table 1). Notably, callus, if at all present, formed at the basal part of the shoot and there was no adventitious shoot formation in the callus.

Rhizogenesis in curly birch under shortened and extended LDCs

Root formation on the shoots cultured *in vitro* under standard conditions (16/8 h LDC) commonly occurs on the 10th-14th days. However, in the extended LDC (48/24 h) treatment adventitious roots appeared as early as on the 7th day in 20% of shoots, and by the 10th day 81% of shoots formed roots (Table 1, Fig. 2). In the control (16/8 h), only 73% of shoots had roots even after 14 days, and the percentage under the shortened (8/4 h) LDC was 64%. For adventitious root formation and for the time until rooting, the most favorable were the conditions of the 48/24 h LDC treatment, where the number of roots after 14 days ranged from 1 to 7, versus 0-5 for the 16/8 and 8/4 h LDCs. Adventitious root length was also greater under the 48/24 h LDC, averaging 17.0 mm (Table 1),

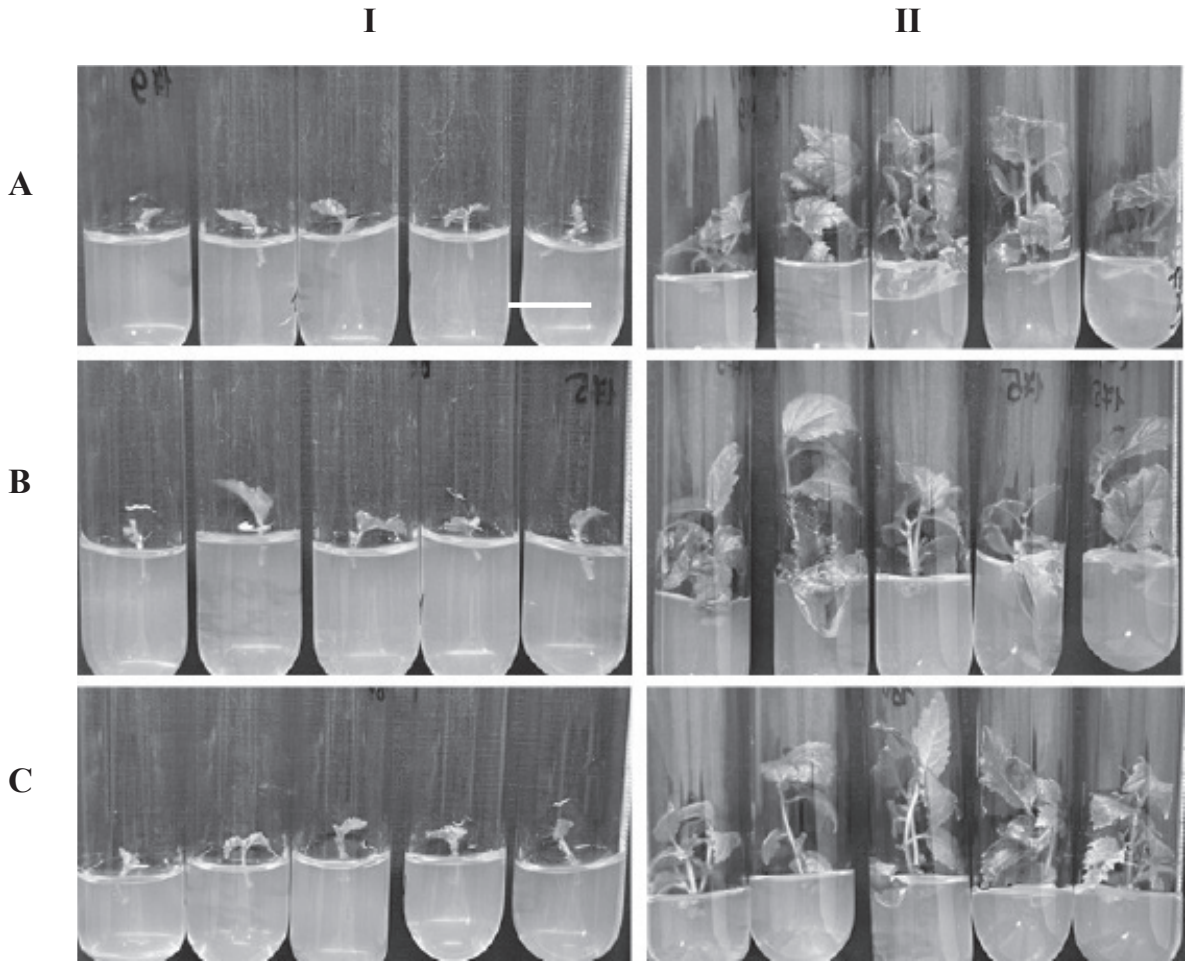


Fig. 1. Growth of axillary shoots in curly birch *in vitro* under different LDCs: A) 16/8 h , B) 8/4 h, C) 48/24 h at the beginning of experiment (I) and after 25 days (scale bar = 1 cm).

Table 1. Growth parameters and rooting of axillary shoots formed under different light-dark cycle conditions *in vitro*.

Parameter	Light-dark cycle (h)		
	16/8	8/4	48/24
	Growth of axillary shoots		
Original shoot length (mm)	13.2 ± 0.7 b	12.5 ± 0.9 b	17.2 ± 0.9 a
Original shoot elongation (mm)	9.3 ± 0.5 b	8.7 ± 0.8 b	12.9 ± 0.9 a
Number of formed shoots	0.9 ± 0.1 a	0.5 ± 0.1 b	0.9 ± 0.1 a
Shoot mass (mg)	90.2 ± 6.5 a	78.3 ± 13.4 b	80.0 ± 7.1 b
Number of leaves	8.7 ± 0.6 a	6.6 ± 0.5 b	8.2 ± 0.6 a
Leaf mass (mg)	70.6 ± 6.4 a	60.4 ± 13.5 b	50.7 ± 6.3 b
Stem mass (mg)	19.6 ± 1.4 b	18.1 ± 2.3 b	29.3 ± 2.9 a
Callus mass (mg)	71.6 ± 5.6 a	46.9 ± 9.9 b	40.0 ± 4.3 b
Total biomass (shoots + callus) (mg)	161.8 ± 7.8 a	115.4 ± 13.9 b	120.0 ± 7.1 b
	Rhizogenesis		
Number of roots	2.4 ± 0.4 b	2.5 ± 0.5 b	3.9 ± 0.5 a
Root length (mm)	14.3 ± 2.2 b	14.0 ± 2.9 b	17.0 ± 2.1 a
Rooted shoots (%)	73.2 ± 3.1 b	64.4 ± 5.1 c	81.6 ± 4.3 a

Means ± standard errors within a row followed by the same letter are not significantly different according Tukey's multiple range test at $p < 0.05$.

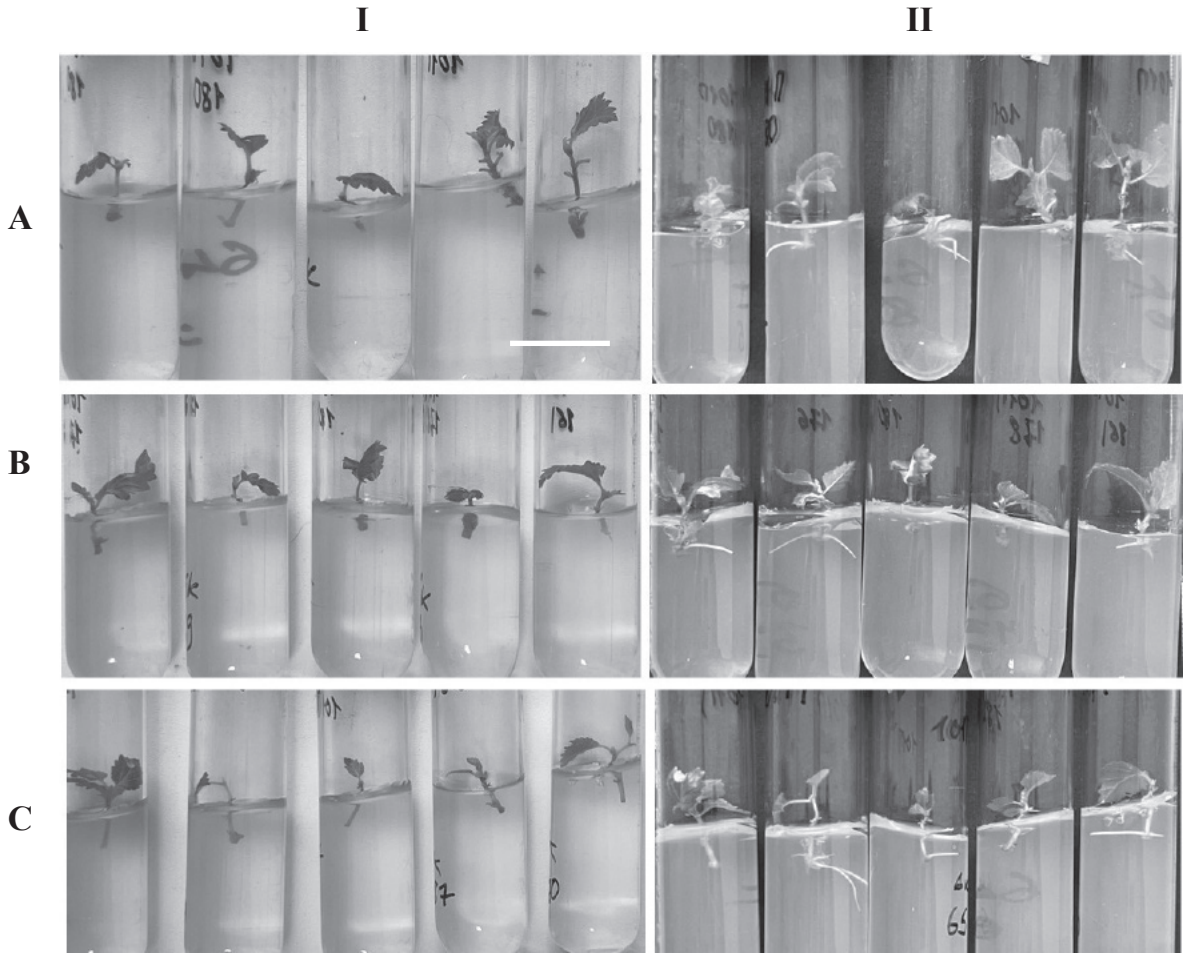


Fig. 2. Rhizogenesis in curly birch under different LDCs: A) 16/8 h, B) 8/4 h, and C) 48/24 h after I) 3 days and II) 10 days after the beginning of experiment (scale bar = 1 cm).

i.e. about 20% more than in the control. However, accelerated root growth is not always desirable, as the fragility of such roots makes them more prone to mechanical damage during *in vitro* to *ex vitro* transfer.

The differences in shoot growth parameters observed under different LDCs during the development of axillary shoots persisted during rhizogenesis. Specifically, the average shoot length was the highest under the 48/24 h LDC (17.6 mm) compared to the 16/8 and 8/4 h LDCs (13.7 and 13.2 mm, respectively). New axillary shoots were formed during rhizogenesis (Table 1).

Acclimatization and growth of regenerated curly birch plants ex vitro in the aftereffect of different LDCs

The survival rate of regenerated plants (Fig. 3) after transfer to *ex vitro* environment was 98% in all treatments including control (data are not shown).

After the transfer to soil, the initial length of the regenerated plants cultured at 48/24 h LDC was 30% greater than in the control (Table 2). After 15 days in soil, the plant length was 1.4 times higher, and after 30 days, it increased nearly six-fold, varying from 9.1 and 9.3 cm to 12.6 cm (for 16/8, 8/4, and 48/24 h LDCs, respectively). However, these differences between plants originally cultured at different LDCs persisted for 15 to 30 days after the transfer to identical-light *ex vitro* conditions and concerned only the variant 48/24. Later on, as the "stimulatory effect" decreased, plant length gradually evened out due to reduced growth rates in the experimental variant, so that by day 45 differences between variants disappeared. After 3 months of growing *ex vitro*, plants reached 45-46 cm (Table 2).

Table 2. Aftereffect of different light-dark cycles on length (cm) of plants *ex vitro*.

Time (days)	Light-dark cycle (h)		
	16/8	8/4	48/24
0	1.6 ± 0.5 b	1.8 ± 0.5 b	2.2 ± 0.6 a
15	6.8 ± 2.2 b	6.4 ± 1.5 b	8.8 ± 2.6 a
30	9.1 ± 2.5 b	9.3 ± 2.3 b	12.6 ± 5.0 a
45	22.6 ± 5.2 a	23.0 ± 4.6 a	22.2 ± 9.0 a
55	29.8 ± 6.0 a	29.9 ± 6.0 a	29.2 ± 11.1 a
60	31.7 ± 6.1 a	32.2 ± 5.8 a	32.2 ± 11.1 a
65	34.1 ± 6.3 a	34.7 ± 5.8 a	34.2 ± 11.2 a
70	35.7 ± 6.1 a	35.9 ± 5.4 a	35.5 ± 11.0 a
90	46.3 ± 7.1 a	45.5 ± 6.7 a	45.8 ± 8.3 a

Means ± standard errors within a row followed by the same letter are not significantly different according Tukey's multiple range test at $p < 0.05$.

DISCUSSION

The axillary shoots formation and rhizogenesis of curly birch were significantly affected by shortened (8/4 h) and extended (48/24 h) LDCs. Abnormal LDCs hindered shoot biomass increment compared to the 16/8 h control. Shortened LDC 8/4 h inhibited the formation of new axillary shoots and the emergence of new leaves more significantly than prolonged 48/24 h LDC (Table 1). This aligns to some extent with known data on growth rate reduction in different poplar species and hybridss under short-day conditions (Eriksson et al. 2015, Maurya and Bhalerao 2017).

The extended LDC stimulated *in vitro* shoot elongation and formation of axillary shoots, which developed 3-4 days earlier than in other LTCs (16/8 and 8/4 h). However, the leaf-to-stem mass ratio under the 48/24 h LDC was half that observed in other LDCs. This was possibly due to the fact that apical meristem cells are mixotrophic in tissue culture and may have additionally utilized sucrose (together with other components) from

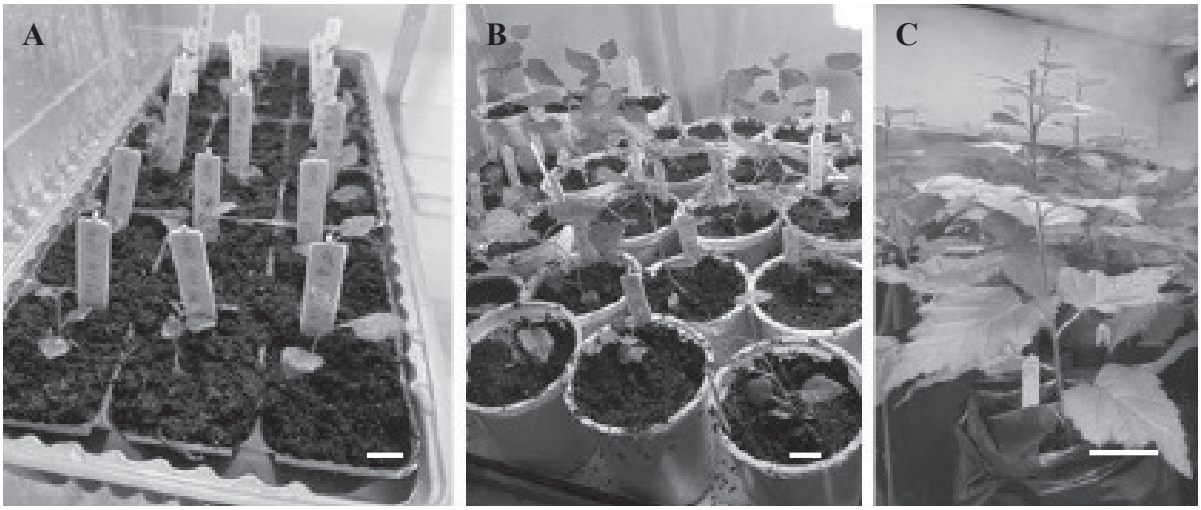


Fig. 3. Acclimatization of curly birch plantlets: A) 6 days, B) 25 days, and C) 65 days from the beginning of experiment in *ex vitro* conditions (scale bar A, B – 1 cm, C – 10 cm).

the nutrient medium for stem tissue formation, increasing stem biomass 1.5-fold compared to other variants (Praveena and Veeresham 2014). We have previously observed a similar positive effect on shoot elongation *in vitro* when axillary shoots were affected by low cadmium concentrations (10^{-6} M) (Vetchinnikova and Titov 2022).

Light affects plant growth by a number of mechanisms. These mechanisms include photosynthesis, the circadian rhythm, and other that are still under study. Shoot growth has lately been shown to be associated with CO (CONSTANS) and FT (FLOWERING LOCUS T) protein expression: when daylight hours are long, CO accumulates, activating the *FT2* gene, whose products, in turn, induce shoot growth. Short days reverse this process: CO protein degrades, *FT* gene expression declines, and shoot growth is terminated (Böhlenius et al. 2006, Azeez and Sane 2015, Maurya and Bhalerao 2017, Romero et al. 2024). A case study of hybrid aspen (*Populus tremula* L. × *tremuloides* Michx) demonstrated the involvement of two signaling pathways in photoperiodic responses: via CO/FT and gibberellin (GA), which regulate shoot elongation (Olsen et al. 2010, Eriksson et al. 2015, Singh et al. 2021), with the following engagement of a complex regulatory network entailing rapid and significant changes in the transcriptome (Chen et al. 2004). A protein additionally identified for *Populus* spp. is HY5 (ELONGATED HYPOCOTYL 5), which the authors (Hajdu et al. 2018, Singh et al. 2021, Gao et al. 2024) believe to suppresses circadian fluctuations of LHY1 and/or LHY2 (cytokinin biosynthesis controllers), thereby activating *FT2* gene expression. It has also been demonstrated that FT2 signaling pathway activation is required for GA pathway activation and, hence, apical shoot growth in long daylight conditions (Gómez-Soto et al. 2022). The activity of the GA20ox1 enzyme, which restrains GA biosynthesis in *Populus* spp., is regulated by daylight duration but can decline under the impact of several shortened LDCs (Singh et al. 2021). The researches are going on to investigate the mechanisms of plant responses to photoperiod changes (Roeber et al. 2022, Gao et al. 2024), which is especially important for woody species the life span of which can be tens or even hundreds of years.

It is possible that the shortened LDC induced an inhibitory effect on the growth of both shoots and roots by mechanisms similar to that described by Gering et al. (1986). Stimulation and acceleration of rhizogenesis in our experiments by extended LDC is similar to some extent to phytochrome-controlled response (Chen et al. 2004) of downy birch seeds, which typically germinate better under long day (20 h) (Lir et al. 1974). The rooting rate and root number under the shortened LDC (8/4) were similar to the control treatment (16/8), which is consistent with the results obtained *in vitro* for most of the fruit tree shoots cultured *in vitro* (fruit tree rootstocks GF 677 (peach × almond hybrid), Mr.S. 2/5 (*Prunus cerasifera* Ehrh.), MM 106 (apple Nothern Spy × Paradise M1) and BA 29 (*Cydonia oblonga* Mill.)) (Morini and Perrone 2006).

Thus, the example of the apical meristem of curly birch shoots *in vitro* demonstrates the high potential of LDC manipulation to influence axillary shoots formation and rhizogenesis performance as well as meristem viability in general. In particular, the extended (48/24 h) LDC augments the number of axillary shoots, promotes their elongation, and stimulates root system formation *in vitro*, enhancing the performance of clonal micropropagation. Its effects persist *ex vitro*, but not more than for one month, which is not so tangible for woody plants with their multi-decade-long life cycles. The shortened (8/4 h) LDC, on the contrary inhibits shoot formation, thus reducing their subculturing frequency during off-seasons and helping save on labor and energy costs (considering time-of-use energy tariffs). A beneficial effect was that non-standard LDCs inhibited callus formation, which is considered undesirable in shoot cultures. Hence, the application of specific LDCs can provide better solutions to the challenges associated with *in vitro* cultivation of curly birch using axillary shoots. This approach prioritizes genetic stability over propagation speed, aligning with conservation objectives for this valuable genotype given its high phenotypic variability in natural populations. This may bring certain economic benefits in the long term, especially when the establishment of artificial stands and/or plantations of this valuable tree species is built up to an industrial scale.

Acknowledgements: The research was carried out using the equipment of the Core Facility of the Karelian Research Centre of the Russian Academy of Sciences. Support for this study was provided by the Russian Science Foundation (RSF, funding number 23–16–00160).

The authors are sincerely grateful to A.G. Padchina, A.I. Stepanova, and I.A. Levkin for their assistance in conducting the experiments.

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